

- (v) Om die temperatuur van die stoom tot  $120^{\circ}\text{C}$  te verhoog, benodig 'n hoeveelheid warmte  $Q_5 = 800\text{ J}$ . Berekken die spesifieke warmtekapasiteit van die stoom.

In order to increase the temperature of the steam to  $120^{\circ}\text{C}$ , heat  $Q_5 = 800\text{ J}$  is needed. Determine the specific heat capacity of steam.

$$Q = cm\Delta T$$

$$800 = c \times 0,01 \times 10$$

$$c = \frac{800}{0,01 \times 10}$$

$$c = 2000\text{ J/kg}^{\circ}\text{C}$$

(2)

- (vi) Hoeveel is die totale warmte wat vir die hele proses nodig was?

How much is the total heat required for the whole process?

$$Q_1 + Q_2 + Q_3 + Q_4 + Q_5 = 1045 + 6680 + 8372 + 45200 + 800$$

$$Q_{\text{Tot}} = 62097\text{ J}$$

(1)

[21]

## VRAAG / QUESTION 2

- (a) Definieer hoeksnelheid  $\omega$  en gee die eenhede daarvan.  
Define angular velocity  $\omega$  and give its units.

$$\omega = \frac{d\theta}{dt}$$

$$[\text{rad/s}]$$

(2)

- (b) (i) 'n Slyp wiel begin draai (vanuit rus) met 'n konstante hoekversnelling  $\alpha = 0,35\text{ rad/s}^2$ . Bepaal die hoekverplasing  $\theta$  wat die wiel in 18 s maak.  
A grindstone starts from rest and has a constant angular acceleration  $\alpha = 0,35\text{ rad/s}^2$ . What is the angular displacement  $\theta$  at  $t = 18\text{ s}$ ?

$$0,35 \times 18^2$$

$$\theta = \frac{113,4}{6}$$

$$= 18,9\text{ rad}$$

$$\omega = \frac{v}{r}$$

$$\alpha = \frac{a}{r}$$

$$\theta = \frac{d}{r}$$

(2)

- (ii) Hoeveel omwentelinge het die wiel gemaak in 18 s?  
Determine the number of revolutions completed in the 18 s.

$$6,3 \times 2\pi = 1,00$$

$$113,4 \div 2\pi = 18,04$$

Wat is die wiel se hoeksnelheid na 18 s?  
What is the wheel's angular velocity at  $t = 18$  s?

$$\alpha = 0,35 \text{ rad/s}$$

$$0,35 \times 18 = 6,3 \text{ rad/s}$$

(2)

- c) 'n Dun ring met massa  $m$  en straal  $R$  roteer om 'n as deur sy middelpunt. Toon aan dat die traagheidsmoment van die hele ring is  $I = mR^2$ .

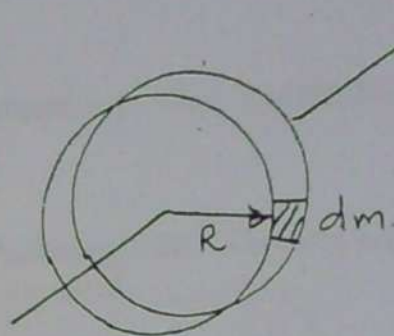
A thin ring has a mass  $m$  and radius  $R$  and it rotates about an axis through its center. Show that the moment of inertia of the whole ring is  $I = mR^2$ .

$$I = \int_{R_1}^{R_2} m dR r^2$$

$$R_1 = 0$$

$$= m (R_2 - R_1)^2$$

$$= m R^2$$

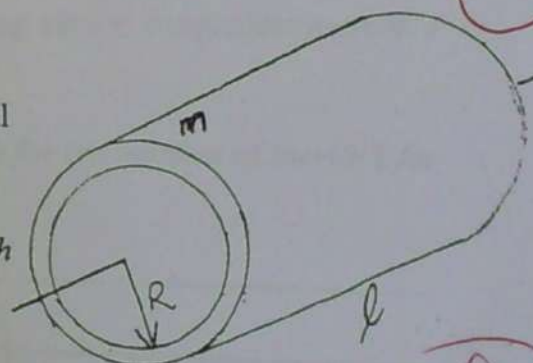


(2)

- (d) Wat is die traagheidsmoment van 'n dun silinder met straal  $R$ , massa  $m$  en lengte  $\ell$ ?

What is the moment of inertia of a thin cylinder with radius  $R$ , mass  $m$  and length  $\ell$ ?

$$I = \frac{1}{2} m R^2$$



(1)

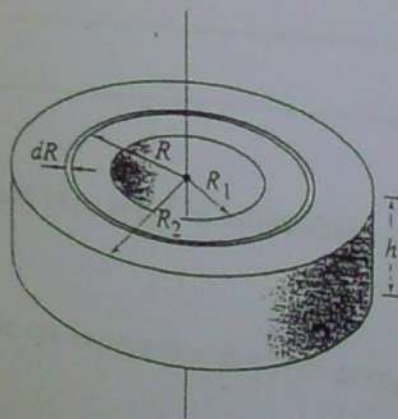
- (e) 'n Sloopwiel het 'n gat in die middel met 'n binne radius  $R_1$ , 'n buiteradius  $R_2$  en 'n massa  $M$ .

A grindstone with a hole in the center has an inner radius  $R_1$ , an outer radius  $R_2$  and a mass  $M$ .

- (i) Skryf 'n uitdrukking neer vir die volume  $V$  van die wiel.  
Give an expression for the volume  $V$  of the wheel.

$$V = \pi R_2^2 h - \pi R_1^2 h$$

$$= \pi h (R_2^2 - R_1^2)$$



(1)

- (ii) Beskou die wiel as opgebou uit dun ringe met dikte  $dR$ . Gee 'n uitdrukking vir die volume  $dV$  van 'n ring met straal  $R$ , dikte  $dR$  en hoogte  $h$ .

The cylinder can be constructed from a number of thin concentric cylindrical rings of thickness  $dR$ . Give an expression for the volume  $dV$  of a ring with radius  $R$ , thickness  $dR$  and height  $h$ .

$$dV = \pi h (R - dR)^2$$

(1)

- (iii) 'n Dun ring met massa  $dm$ , volume  $dV$  en dikte  $dR$  se digtheid is  $\rho$ . Gee 'n uitdrukking vir  $dm$  in terme van  $\rho$ .

The mass of a thin ring is  $dm$ , its volume is  $dV$  and its thickness is  $dR$ . The ring has a density  $\rho$ . Give an expression for  $dm$  in terms of  $\rho$ .

$$2\pi R dR \times dm = \rho$$

$$2\pi R dR \times dm = \rho$$

$$dm = \frac{\rho}{2\pi R dR}$$

$$dm = \frac{d\rho}{2\pi R dR}$$

$$1 dm = \frac{d\rho}{2\pi R dR}$$

$$2\pi R dR \times h \quad (2)$$

- (iv) Met bostaande uitdrukking vir  $dm$ , herlei 'n uitdrukking vir die traagheidsmoment  $I$  vir die wiel in terme van  $R_1$  en  $R_2$  en  $\rho$

With the expression for  $dm$  above, derive an expression for the moment of inertia  $I$  for the whole wheel in terms of  $R_1$  and  $R_2$  and  $\rho$ .

$$I = \frac{1}{2} \rho \frac{(R_2 - R_1)^2}{2\pi (R_2 - R_1)}$$

$$I = \frac{\rho \cdot (R_2 - R_1) \cdot h}{2 \cdot 2\pi}$$

(5)

- (v) Indien die totale massa van die wiel  $M$  is en sy digtheid is  $\rho$ , skryf 'n uitdrukking vir  $\rho$  in terme van  $M$ ,  $R_1$  en  $R_2$ .

The wheel has a total mass  $M$  and density  $\rho$ . Write an expression for  $\rho$  in terms of  $M$ ,  $R_1$  and  $R_2$ .

$$\rho = \frac{M \times (R_2 + R_1) \cdot 2\pi \cdot h}{(R_2 - R_1)^2 \cdot 2\pi \cdot h}$$

$$\rho = \frac{M}{(R_2 - R_1)^2 \cdot 2\pi \cdot h}$$

(1)

- (ii) Beskou die wiel as opgebou uit dun ringe met dikte  $dR$ . Gee 'n uitdrukking vir die volume  $dV$  van 'n ring met straal  $R$ , dikte  $dR$  en hoogte  $h$ .

The cylinder can be constructed from a number of thin concentric cylindrical rings of thickness  $dR$ . Give an expression for the volume  $dV$  of a ring with radius  $R$ , thickness  $dR$  and height  $h$ .

$$dV = \pi h (R - dR)^2$$

(1)

- (iii) 'n Dun ring met massa  $dm$ , volume  $dV$  en dikte  $dR$  se digtheid is  $\rho$ . Gee 'n uitdrukking vir  $dm$  in terme van  $\rho$ .

The mass of a thin ring is  $dm$ , its volume is  $dV$  and its thickness is  $dR$ . The ring has a density  $\rho$ . Give an expression for  $dm$  in terms of  $\rho$ .

$$2\pi dR \times dm = \rho$$

$$2\pi dR \times dm = \rho$$

$$dm = \frac{\rho}{2\pi dR}$$

$$dm = \frac{\rho}{2\pi dR}$$

$$2 dm = \frac{\rho}{\pi dR}$$

$$2\pi dR \times h \cdot \rho = dm$$

(2)

- (iv) Met bostaande uitdrukking vir  $dm$ , herlei 'n uitdrukking vir die traagheidsmoment  $I$  vir die wiel in terme van  $R_1$  en  $R_2$  en  $\rho$ .

With the expression for  $dm$  above, derive an expression for the moment of inertia  $I$  for the whole wheel in terms of  $R_1$  and  $R_2$  and  $\rho$ .

$$I = \frac{1}{2} \rho \frac{(R_2 - R_1)^2}{2\pi (R_2 - R_1)}$$

$$I = \frac{\rho (R_2 - R_1) \cdot bh}{2\pi \cdot 2\pi}$$

(5)

- (v) Indien die totale massa van die wiel  $M$  is en sy digtheid is  $\rho$ , skryf 'n uitdrukking vir  $\rho$  in terme van  $M$ ,  $R_1$  en  $R_2$ .

The wheel has a total mass  $M$  and density  $\rho$ . Write an expression for  $\rho$  in terms of  $M$ ,  $R_1$  and  $R_2$ .

$$\rho = \frac{M \times (R_2 + R_1) \cdot 2\pi \cdot h}{(R_2 - R_1)^2 \cdot 2\pi \cdot h}$$

$$\rho = \frac{M}{(R_2 - R_1)^2 \cdot 2\pi \cdot h}$$

(1)

- (vi) Met  $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$ , toon aan dat  
 $I = \frac{1}{2} M (R_2^2 + R_1^2)$ .

With  $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$ , show that  
 $I = \frac{1}{2} M (R_2^2 + R_1^2)$ .

$$\begin{aligned}
 I &= \frac{\rho (R_2^4 - R_1^4)}{2 \cdot 2\pi} \\
 &= \frac{\rho (R_2^2 - R_1^2)(R_2^2 + R_1^2)}{2 \cdot 2\pi} \cdot 2\pi \cdot h \\
 &= \frac{\rho (R_2^2 - R_1^2)(R_2^2 + R_1^2)}{2} \cdot h \\
 I &= \frac{1}{2} M (R_2^2 + R_1^2)
 \end{aligned}$$

- (f) In die figuur word die sy-aansig van 'n motorkar se band getoon. Dit het twee sykante met 'n dikte van  $d = 0,635$  cm. Die dikte van die loopvlak is 2,50 cm en is 20 cm breed. Die digtheid van rubber is  $\rho = 1,10 \times 10^3 \text{ kg/m}^3$ .

The figure shows a side view of a car tyre. It has two side walls of thickness  $d = 0,635$  cm. The thickness of the tread wall is 2.50 cm and it is 20 cm wide. The density of rubber is  $\rho = 1,10 \times 10^3 \text{ kg/m}^3$ .



- (i) Toon die volledige berekening om aan te toon dat die traagheidsmoment van die twee sykante  $0,1736 \text{ kg m}^2$  is indien die traagheidsmoment van 'n dun skyf gegee word deur  $\frac{1}{2} MR^2$ .

Show step by step the calculations to determine that the moment of inertia of the two sides is  $0,1736 \text{ kg m}^2$  if the moment of inertia of a thin disk is given by  $\frac{1}{2} MR^2$ .

$$\begin{aligned}
 I &= \frac{1}{2} M R^2 \\
 &= \frac{1}{2} \frac{\rho}{0,635 \times 10^{-3}} (0,305 - 0,165)^2 \\
 &= \frac{1}{2} \frac{1100}{0,635 \times (0,305 - 0,165)} (0,305 - 0,165)^2 \\
 &= \frac{1}{2} \times 17,715 \times (0,305 - 0,165)^2 \\
 &= 0,1736 \text{ kg m}^2
 \end{aligned}$$

- (ii) Toon volledig dat die traagheidsmoment van die loopvlak alleen  $1,10776 \text{ kg m}^2$  is.

Show that the moment of inertia of the tread only is  $1,10776 \text{ kg m}^2$ .

$$I = \frac{1}{2} MR^2$$

$$= \frac{1}{2} \frac{P}{0,12 \times 10,1776 \times 0,12 \times 10,1776} \times 0,12^2$$

$$= 1,107698$$

$$\approx 1,10776$$

(4)

- (iii) Wat is die totale traagheidsmoment van die hele band?

What is the total moment of inertia of the whole tyre?

$$0,1736 \times 2 + 1,10776 = 1,45496 \text{ kg m}^2$$

Sidech tyres + 6 of them of 6,25 kg

$$= (0,1736 \times 2) + 1,10776 + 1,45496 \text{ kg m}^2$$

(2)

- (g) Indien die wiel roteer met 'n hoeksnelheid  $\omega = 600$  omwentelings per minuut, bepaal die kinetiese energie wat in die wiel gestoor is.

If the wheel is rotating with an angular velocity  $\omega = 600$  revolutions per minute, determine the kinetic energy stored in the wheel.

$$E_k = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times 1,45496 \times 22,0468^2$$

$$= 3,468 \times 10^4 \text{ J}$$

$$m = 1,45496$$

$$(0,33 \times 0,12)$$

$$= 22,0468 \dots \times 2$$

(3)

## VRAAG / QUESTION 3

- (a) 'n Voorwerp met massa  $m$  beweeg in 'n sirkel (straal  $r$ ) met 'n konstante spoed  $v$ . Skryf 'n uitdrukking neer vir die grootte van die sentripetale krag  $F$ .

An object with mass  $m$  has a circular motion (radius  $r$ ) at a constant speed  $v$ . Give an expression for the magnitude of the centripetal force  $F$ .

$$F = \frac{mv^2}{r}$$

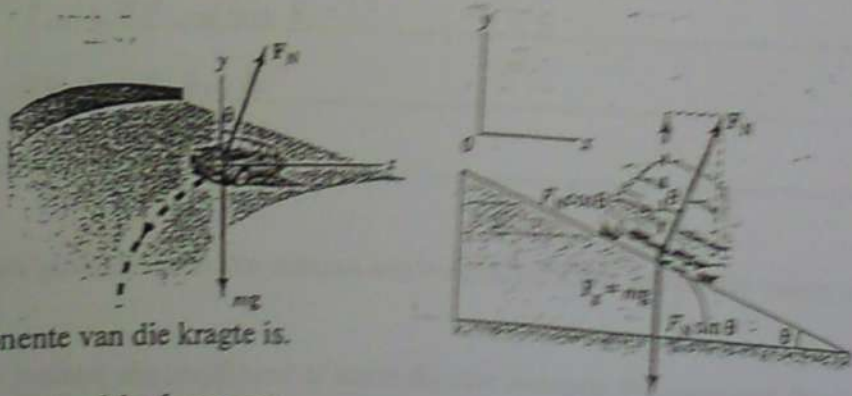
- (b) Met  $\mu_s$  die statiese wrywingskoeffisiënt, skryf 'n uitdrukking neer vir die statiese wrywingskrag  $f_s$ .

The coefficient for static friction is  $\mu_s$ . Give an expression for the force of static friction  $f_s$ .

$$f_s = \mu_s \times F_n$$

- (c) 'n Motorkar gaan om 'n draai in die pad waarvan die radius  $R$  m is. Die pad is skuins gemaak met 'n hoek  $\theta$ . Weens 'n laag sneeu op die pad is daar geen wrywing tussen die bande en die pad nie.

An automobile makes a turn whose radius is  $R$  m. The road is banked at an angle of  $\theta$ . Due to snow on the road there is no friction between the tyres and the road.



- (i) Die x-komponente van die kragte is.

The x-components of the forces are.

$$\Sigma F_x = F_n \sin \theta - f_s$$

- (ii) Die y-komponente van die kragte is.

The y-components of the forces are.

$$\Sigma F_y = F_n \cos \theta - mg$$

- (iii) Toon aan dat die spoed van die kar gegee word deur  $v = (gR \tan \theta)^{1/2}$ .

Show that the speed the driver should take the curve is given by  $v = (gR \tan \theta)^{1/2}$ .

$$V = \frac{F}{R} = \frac{F_n \sin \theta - f_s}{r}$$

$$= \frac{mg \sin \theta - f_s}{r}$$

$$= (gR \tan \theta)^{1/2}$$

(3)

- (iv) Met  $R = 320$  m en  $\theta = 5,1^\circ$ , bepaal die spoed van die voertuig (in  $\text{km h}^{-1}$ ) sodat die voertuig nie afgly nie.

With  $R = 320$  m and  $\theta = 5,1^\circ$ , determine the speed (in  $\text{km h}^{-1}$ ) the driver should take the curve to avoid sliding off the road.

$$v = (gR \tan \theta)^{1/2}$$

$$= (-9,8 \times 320 \times \tan 5,1)^{1/2}$$

$$= 87,64 \text{ m/s} \text{ kmh}^{-1}$$

$$= 26,26 \text{ m/s}$$

(3)

- (v) Nadat die sneeu gesmelt het is die statiese wrywingskoëffisiënt tussen die bande en die pad 0,8.

After the snow melted, the coefficient of static friction between the tyres and the road is 0,8.

x-komponente / x-components

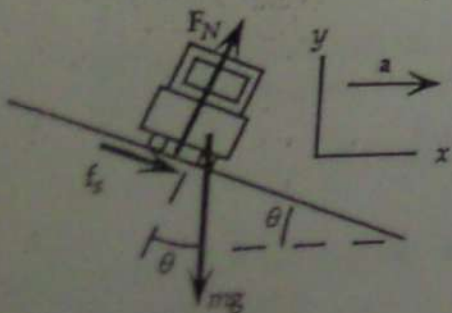
$$\Sigma F_x = mg \sin \theta + f_s \sin \theta$$

(1)

y-komponente / y-components

$$\Sigma F_y = mg - F_n \cos \theta$$

(1)



(vi) Bepaal die maksimum speed (in  $\text{km h}^{-1}$ ) wat die voertuig om die draai kan gaan.

Determine the maximum speed (in  $\text{km h}^{-1}$ ) that the motorist can take the curve.

$$\begin{aligned} \sum F_{\text{net}} &= F_n \sin \theta + f_s \sin \theta \\ mg &= F_n \sin \theta + f_s \sin \theta \\ m(9,8) &= F_n \sin(5,1) + f_s \sin(5,1) \\ 2493,02 &= F_n 0,0888 + f_s 0,0888 \\ 28074,58 &= F_n + f_s \\ 87,66 \times 320 &= F_n + 0,8 \times F_n \\ &= 28044,8 \text{ N} \\ 28074,58 &= F_n (1 + 0,8) \\ F_n &= 15596,98934 \\ 28044,8 \text{ N} \times \sin(5,1) &= F_n \sin(5,1) \\ F_n &= 2493,02 \text{ N} \\ v &= \frac{F}{r} = \frac{15596,98934}{320} = 48,74 \text{ km h}^{-1} \end{aligned}$$

#### VRAAG / QUESTION 4

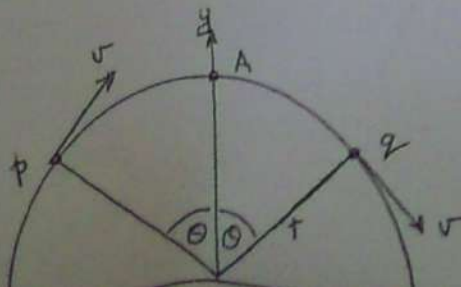
- (a) 'n Voorwerp beweeg langs die x-as. Op tydstep  $t_1$  het dit 'n snelheid  $v_1$ , waarna dit versnel om op tydstep  $t_2$  'n snelheid  $v_2$  te behaal. Gee 'n uitdrukking vir die gemiddelde versnelling  $\langle a_x \rangle$  langs die x-as.

An object moves along the x-axis. It accelerates along the x-axis from velocity  $v_1$  (at  $t_1$ ) to  $v_2$  (at  $t_2$ ). Give an expression for the average acceleration  $\langle a_x \rangle$  along the x-axis.

$$a_x = \frac{(v_2 + v_1)}{2} t$$

- (b) 'n Voorwerp beweeg met 'n konstante speed  $v$  in 'n sirkelbaan, (in 'n horisontale plat vlak) soos in die figuur.

An object moves with a constant speed  $v$  in a circle, (in a horizontal flat plane) as shown in the figure.



- (i) Bereken die gemiddelde versnelling in die x-as rigting.

Calculate the average acceleration in the x-direction.

$$a = \frac{v^2}{r} \frac{\cos \theta}{\theta}$$

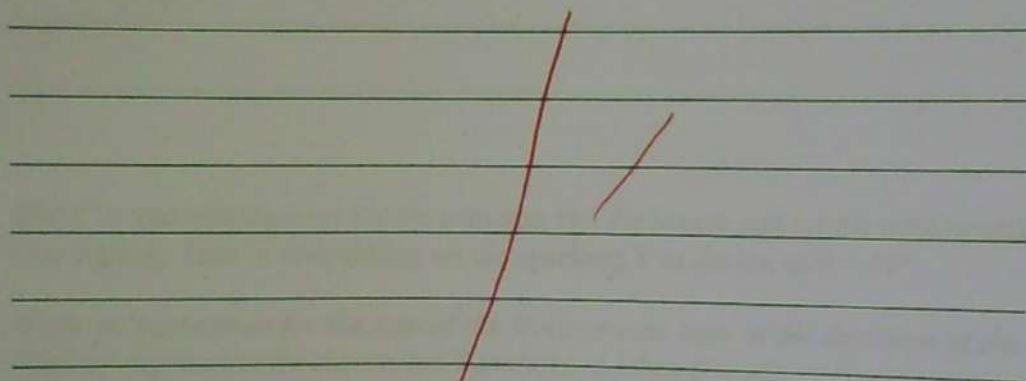
$$= \frac{0^2}{r} \frac{\cos \theta}{\theta} = \frac{\cos \theta}{r \theta} = +10825 \text{ m/s}^2$$

(3)

- (ii) Toon aan dat die gemiddelde versnelling in die y-as rigting gegee word deur:

Show that the average acceleration in the direction of the y-axis is given by:

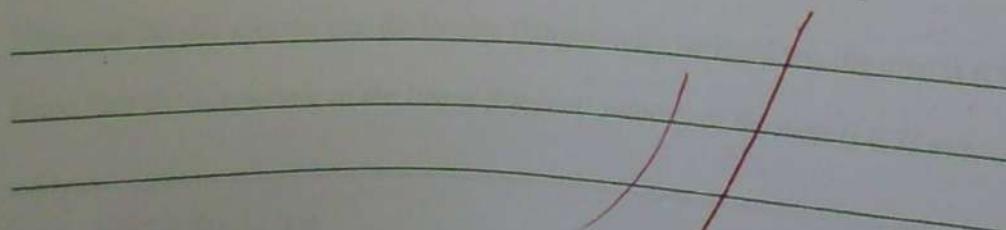
$$\langle a_y \rangle = -\frac{v^2}{r} \frac{\sin \theta}{\theta}$$



(3)

- (iii) Toon aan dat die grootte van die versnelling by A gegee word deur  $|\vec{a}| = \frac{v^2}{r}$ .

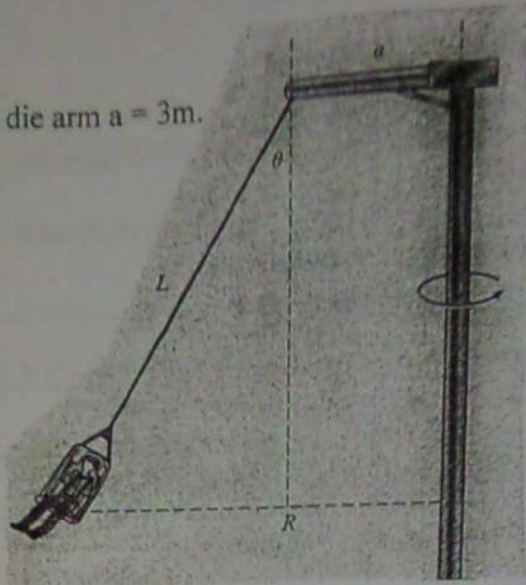
Show that the magnitude of the acceleration at A is given by  $|\vec{a}| = \frac{v^2}{r}$ .



(2)

- (c) 'n Roterende swaai het 'n tou met lengte  $L = 10$  m en die arm  $a = 3$  m.

A rotating swing has a rope with length  $L = 10$  m and the arm length is  $a = 3$  m.



- (i) Toon aan dat die radius  $R$  van die roterende man is 8 m. en  $\theta = 30^\circ$

Show that the radius  $R$  of the rotating man is 8 m. and  $\theta = 30^\circ$

$$\begin{aligned} \cancel{3 + 10 \sin \theta} & \quad 3 + 10 \sin \theta = R \\ \cancel{3 + 10 \sin 30} & \quad 3 + 10 \sin(30) = R \\ \cancel{8} & \quad R = 8 \end{aligned}$$

(2)

- (ii) Skryf 'n uitdrukking neer vir die som van van die kragte wat op die man inwerk in die y-as rigting. Gee 'n uitdrukking vir die spankrag  $T$  in die tou as  $\theta = 30^\circ$ .

Write an expression for the sum of the forces on the man in the direction of the y-axis. Find an expression for the tension  $T$  in the rope when  $\theta = 30^\circ$ .

$$\begin{aligned} T &= F_y \\ T &= F_r \sin 30 \\ T &= T_r \sin 30 \end{aligned}$$

(2)

- (iii) Toon aan dat die grootte van die lineêre (tangensiale) snelheid van die man is 6,73 m/s.

Show that the magnitude of the linear (tangent) velocity of the man is 6,73 m/s.

$$\begin{aligned} 1,673 \times 8 \sin 30 \\ = 6,73 \text{ m/s} \\ \therefore 6,73 \text{ m/s} \end{aligned}$$

(3)

- (iv) Indien die man se massa 97,21 kg is, bepaal die spankrag in die tou.

If the man has a mass of 97,21 kg, determine the tension in the rope.

$$T = F_n \cos \theta + F_n \sin \theta$$

$$= (9,8 \times 97,21) \cos(30) + (9,8 \times 97,21) \sin(30)$$

$$T = 1301,36 \text{ N}$$

(2)

- (v) Bepaal die frekwensie  $f$  waarteen die man swaai as  $\theta = 30^\circ$ .

Determine the frequency  $f$  of the swinging man when  $\theta = 30^\circ$ .

$$T = \frac{1}{f} \sqrt{\frac{L}{g}}$$

$$= \frac{1}{f} \sqrt{\frac{9,8}{g}}$$

$$= 0,176 \text{ Hz}$$

$$= 0,176 \text{ Hz}$$

(3)

- (vi) Wanneer die frekwensie waarteen die man draai verhoog word tot  $f = 0,257 \text{ Hz}$ , dan swaai hy in 'n horisontale vlak met die bokant van die paal. Bereken nou die lineêre snelheid van die man.

When the frequency of the man is increased to  $f = 0,257 \text{ Hz}$ , he is rotating in a horizontal plane with to top of the pole. Calculate the linear velocity of the man.

(3)

- (vii) Wat is die minimum spanning wat die arm moet kan hanteer?

What is the minimum tension that the arm <sup>must</sup> handle?

$$T = F_n \cos \theta + F_n \sin \theta$$

$$= (9,8 \times 97,21) \cos 0 + (9,8 \times 97,21) \sin(90)$$

$$= 1804,133 \text{ N}$$

(2)

- (iv) Indien die man se massa 97,21 kg is, bepaal die spankrag in die tou.

If the man has a mass of 97,21 kg, determine the tension in the rope.

$$T = F_n \cos \theta + F_g \sin \theta$$

$$= (9,8 \times 97,21) \cos(30) + (9,8 \times 97,21) \sin(30)$$

$$T = 1301,36 \text{ N}$$

(2)

- (v) Bepaal die frekwensie  $f$  waartoe die man swaai as  $\theta = 30^\circ$ .

Determine the frequency  $f$  of the swinging man when  $\theta = 30^\circ$ .

$$T = \frac{1}{f} \sqrt{\frac{L}{g}}$$

$$= \frac{1}{f} \sqrt{\frac{9,8}{g}}$$

$$= 0,176 \text{ Hz}$$

$$= 0,176 \text{ Hz}$$

(3)

- (vi) Wanneer die frekwensie waartoe die man draai verhoog word tot  $f = 0,257 \text{ Hz}$ , dan swaai hy in 'n horisontale vlak met die bokant van die paal. Bereken nou die lineêre snelheid van die man.

When the frequency of the man is increased to  $f = 0,257 \text{ Hz}$ , he is rotating in a horizontal plane with to top of the pole. Calculate the linear velocity of the man.

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What is the minimum tension that the arm <sup>must</sup> handle?

$$T = F_n \cos \theta + F_g \sin \theta$$

$$= (9,8 \times 97,21) \cos 0 + (9,8 \times 97,21) \sin(90)$$

$$= 1804,33 \text{ N}$$

(2)

- (viii) Hoeveel keer moet die tou sterker wees om horisontaal te swaai as wat dit moet wees om teen  $30^\circ$  te swaai.

How many times should the rope be stronger to swing horizontally than it was necessary to swing at  $30^\circ$ .

$$T_{30} = 1301,34 \quad T_{90} = 1804,33$$

$$T_{90} = T_{30}$$

$$= 1,386 \text{ times greater}$$

(1)

[27]

### VRAAG / QUESTION 5

- (a) Definieer momentum van 'n voorwerp en gee die eenheid waarin dit gemeet word.

Define momentum of an object and give its unit.

$$p = mv \quad \text{kg m/s}$$

(2)

- (b) Die aanvanklike totale momentum van 'n voorwerp is  $\vec{p}_i$  en dan verdeel dit in twee stukke sodat die finale momenta  $\vec{p}_{f1}$  en  $\vec{p}_{f2}$  is. Uit hierdie gegewens, skryf die wet van behoud van momentum neer.

The initial momentum of an object is  $\vec{p}_i$ . Then it <sup>divides</sup> divides into two pieces and the final momenta is  $\vec{p}_{f1}$  and  $\vec{p}_{f2}$ . From this, write the law of conservation of momentum. (kinetic energy is conserved.)

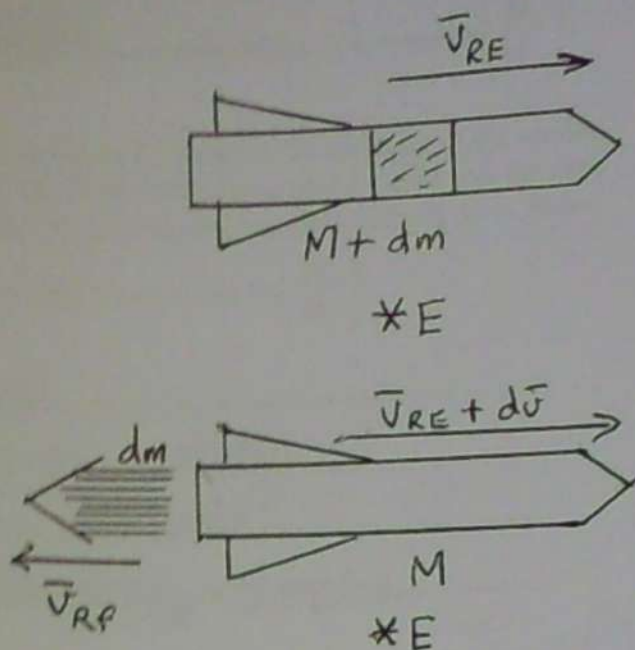
$$\vec{p}_i = \vec{p}_{f1} + \vec{p}_{f2} \quad \text{total before = total after}$$

$$\text{Total momentum before} = \text{Total momentum after collision}$$

(1)

- (e) 'n Vuurpyl ("Rocket") het 'n massa  $M$  en is gelaai met brandstof met massa  $dm$ . Dit beweeg relatief tot die aarde ( $E$ ) met snelheid  $\vec{v}_{RE}$ . Vir 'n kort tyd  $\Delta t$  spuit die vuurpyl brandstof met massa  $dm$  en snelheid  $\vec{v}_{RF}$  uit. Nou is die vuurpyl se massa  $M$  en sy snelheid vermeerde na  $\vec{v}_{RE} + d\vec{v}$ . Hieruit volg dat  $\vec{v}_{RE} = \vec{v}_{RF} + \vec{v}_{RE}$ .

A rocket has a mass  $M$  and is loaded with fuel of mass  $dm$ . It moves relative to the earth ( $E$ ) with velocity  $\vec{v}_{RE}$ . In a short interval  $\Delta t$  the rocket ejects fuel of mass  $dm$  at a velocity  $\vec{v}_{RF}$ . Now the mass of the rocket is  $M$  and its velocity changed to  $\vec{v}_{RE} + d\vec{v}$ . From above it follows that  $\vec{v}_{RE} = \vec{v}_{RF} + \vec{v}_{RE}$ .



- (i) Uit die wet van behoud van momentum (relatief tot die aarde  $E$ ), toon aan dat  $M dv = v_{RF} dm$ .

From the law of conservation of momentum (relative to the earth  $E$ ) show that  $M dv = v_{RF} dm$ .

|                                |   |                |
|--------------------------------|---|----------------|
| Total p before = Total p after |   | $m_1 = M$      |
| $m_1 v_1$                      | = | $m_2 v_2$      |
| $\therefore M dv$              | = | $dm v_{RF}$    |
|                                |   | $v_L = v_{RF}$ |

(3)

- (ii) Die massa brandstof ( $dm$ ) wat uitgelaat word verteenwoordig 'n afname in die vuurpyl se massa met  $dM$ , sodat  $dm = -dM$ . Indien die begin-massa van die vuurpyl en

brandstof  $M_i$  is, is sy beginsnelheid  $v_i$ . Die finale massa is  $M_f$  en die finale snelheid  $v_f$ .  
Toon aan dat die verandering in die snelheid van die vuurpyl is  $v_f - v_i = v_{rel} \ln \left( \frac{M_i}{M_f} \right)$ .

The mass of fuel ( $dm$ ) which is ejected represents a decrease in the mass of the rocket with an amount  $dM$ . Therefore,  $dm = -dM$ . If the initial mass of the rocket and the fuel is  $M_i$ , its initial velocity is  $v_i$ . The final mass is  $M_f$  and the final velocity is  $v_f$ . Show that the change in the velocity of the rocket is  $v_f - v_i = v_{rel} \ln \left( \frac{M_i}{M_f} \right)$ .

$$v_f M_f = v_i M_i + v_{rel} dM$$

$$v_i M_i + v_{rel} dm = v_f M_f + v_{rel} dM$$

$$v_i (M_i - dM) = v_f (M_f + dM)$$

$$v_f - v_i = v_{rel} \frac{M_i - dM}{M_f + dM} \quad \text{subtracting the two equations}$$

$$v_f - v_i = v_{rel} \ln \left( \frac{M_i}{M_f} \right)$$

(3)

- (iii) 'n Vuurpyl het 'n spoed van  $3 \times 10^3$  m/s relatief tot die aarde. Wanneer die vuurpyl aangeskakel word is die snelheid van die brandstof  $5 \times 10^3$  m/s relatief tot die vuurpyl. Bereken die spoed van die vuurpyl wanneer sy massa verminder het tot die helfte van sy massa voor ontbranding.

A rocket is moving at a speed of  $3 \times 10^3$  m/s relative to the earth. When its engines are turned on the velocity of the fuel is  $5 \times 10^3$  m/s relative to the rocket. Calculate the speed of the rocket once the rocket's mass is reduced to half of its mass before ignition.

$$p_{\text{before}} = p_{\text{after}}$$

$$3000 M = v_f (M - dm) + 5000 dm$$

$$v_f = \frac{3000 M}{M - dm + 5000 dm}$$

$$v_f = \frac{3000 M}{(M - dm) + 5000 dm}$$

(4)

- (iv) Wanneer die vuurpyl aangeskakel word is die krag  $F = 2,5 \times 10^5 \text{ N}$ . Indien die vuurpyl se massa  $1428 \text{ kg}$  is, bereken die tyd  $\Delta t$  wat nodig is om sy snelheid te verander tot  $6,5 \times 10^3 \text{ m/s}$ .

When the rocket is ignited, it delivers a force  $F = 2,5 \times 10^5 \text{ N}$ . With the mass of the rocket  $1428 \text{ kg}$ , calculate the time  $\Delta t$  necessary to change its speed to  $6,5 \times 10^3 \text{ m/s}$ .

$$F = ma$$

$$v_f - v_i = v_{af} \ln \left( \frac{m_i}{m_f} \right)$$

$$2,5 \times 10^5 = 1428 a$$

$$a = \frac{2,5 \times 10^5}{1428}$$

$$v_f^2 = v_i^2 + a t$$

$$(6,5 \times 10^3)^2 = (3 \times 10^3)^2 + 175,1438 \dots t$$

$$a = 175,14385465 \text{ m/s}^2$$

$$t = \frac{(6,5 \times 10^3)^2 - (3 \times 10^3)^2}{175,1438 \dots}$$

$$t = 19,95$$

$$\therefore t = 20 \text{ s}$$

(3)

[16]